

Attention-Sink Signatures Dissociate During Vision–Language Pretraining

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Abstract. Attention sinks, token positions that draw heavy attention while contributing little, are a widely studied target of interventions in language models and are often linked, though not causally, to hallucinations in vision-language models. A sink is defined by attention concentration (Sink^e_1), but in text LMs it reliably arrives with two companions: a drained value norm (v-ratio) and massive activations (h-ratio) at the same position. We measure these three signatures during multimodal pretraining: a 222M vision-language model with a randomly initialized decoder, position 0 being an image token, no BOS, trained under four levers: softmax attention, output-gated softmax, unnormalized sigmoid attention, and initialization from a pretrained text LM. We find that the signatures come apart, across 2-3 seeds, depending on which train-time lever was applied. Most notably: on a 1B fresh-token run (2.39 effective visual epochs, single seed), the massive-activation proxy more than doubles while attention concentration stays at exactly zero at every threshold we test. Value-norm drain emerges as a third axis, apart from the massive-activation vs. sink dissociations known from text-only models.

1. Introduction

Decoder-only transformers pick up a habit early in training. A large share of attention goes to the first token or tokens of a sequence, whatever they contain. Streaming inference depends on that habit [1], the activation outliers that ride along with it make quantization harder [2], and a survey now organizes a subfield around interpreting and removing it [3].

The sink does not arrive alone. Three measurements move together in text language models. Attention concentrates on the sink token. The value vector there drops to a near-zero norm, called value-state drain [4]. The residual stream there grows an abnormally large norm, called a massive activation [2, 5]. These effects settle on the same few tokens [6] and appear near step 1000 [7], and the field reads the co-occurrence as permission to treat all three as facets of one attention-sink phenomenon.

Whether they really are one phenomenon is contested. One line argues for causal unity. Massive activations mathematically require representational compression, and ablating them removes both compression valleys and sink formation [7]. A related view holds that outlier-driven rescaling by attention and residual sinks is *essential* to stable training [8]. Two recent studies pull the other way. Each separates a pair of the signatures by intervention, one through the normalization scheme [9] and one through value scale [10]. Both are text-only, and each separates at most two of the three.

Which signature a study measures decides what it can conclude. If the three move independently, a lever that clears concentration while leaving the value path alone reads as a fix under one metric and a failure under the next, and two papers can disagree while both are right. Sink-like attention has also been tied to hallucination and weak visual grounding in deployed vision–language models [13–16], so a mitigation credited with removing the sink may leave the signature that mattered untouched. We measure all three at once, throughout training, and test no downstream behavior.

Vision–language models suit the question, because the multimodal setting changes what position 0 *is*. Our sequence starts with 49 image tokens and has no BOS token, so the candidate sink token is a visual token, without the special-token machinery that text-LM sink accounts lean on. The multimodal sink studies we know analyze mostly frozen, already-trained backbones at inference time [11, 12], and the one training-time view follows a single magnitude rather than three signatures (§4). Telling whether the signatures arrive together, in sequence, or independently takes a randomly initialized decoder with all three logged separately from step 0.

We work at small scale by choice. The model is a 222M-parameter nanoVLM [17], where a pre-trained SigLIP-B/16 encoder [18] feeds a randomly initialized decoder with the SmolLM2-135M architecture [19]. Four levers each target one sink-relevant mechanism (§2): standard softmax attention (*baseline*), a Qiu-style G1 output gate in our zero-initialized variant [20] (*glgate*), unnormalized sigmoid attention, which removes the normalization sinks are argued to come from [6] (*sigmoid*), and initialization from the pretrained SmolLM2 text model (*textinit*). A validated probe logs all three signatures every 100 steps. The four-arm comparison reuses a small image pool, so we re-test the central negative result under low repetition, on a fresh stream of one billion tokens (*RF*, 2.39 effective visual epochs).

Contributions.

1. **Four levers, four corners ($n = 2-3$ seeds/arm).** The arms reach four different corners of the three-signature space, no two sharing a triple. The value-norm ratio alone is strongly drained, mildly drained, or amplified, depending on the lever (Fig. 1, Table 1). The arms are not a factorial design, so we report intervention-associated profiles, not isolated causal effects.
2. **Low-repetition decoupling at 1B tokens ($n = 1$).** On a fresh stream at 2.39 effective visual epochs, with training healthy throughout (§2), the massive-activation proxy rises from an h-ratio of 1.43 to 3.22, about $2.3\times$, while concentration stays at exactly zero and no head ever crosses the sink threshold (§3.2, §5).
3. **No consistent-sign head-level relationship.** The per-head correlation between concentration and value-norm flips sign across arms at seed 0 ($+0.76$ baseline $\rightarrow -0.79$ textinit, pooled -0.20 , over 90 KV groups per arm), reported descriptively (§3.3).

Decoupling itself is not new [9, 10] and we do not claim it. The new part is the conjunction. We track all three signatures jointly, under randomly initialized decoders in a multimodal model, and add value-norm drain as a third axis.

2. Setup

Model and token layout. All runs use a 222M-parameter nanoVLM [17]. A pretrained, trainable SigLIP-B/16 vision encoder [18] feeds a decoder with the SmolLM2-135M architecture [19] through a learned modality projector. The decoder has 30 layers of grouped-query attention, with 9 query heads per layer sharing 3 KV heads. That gives 270 (layer, query-head) pairs but only 90 (layer, KV-group) value projections, a distinction that matters in §3.3. It trains **from random initialization** in every arm except *textinit*, the designed exception, which imports first-position structure from text pretraining (§3.1). Each sequence holds 49 image tokens as a causal prefix, then 79 left-padded text tokens, 128 in all. **Position 0 is the first image token, and there is no BOS token** (§1).

Arms. Four training levers, each targeting one sink-relevant mechanism. Everything else stays byte-identical across arms.

arm	attention	LM init	ViT init	lever precedent
<i>baseline</i>	softmax	random	pretrained	—
<i>glgate</i>	softmax + elementwise σ -gate (zero-init, post-SDPA)	random	pretrained	G1 gating [20]
<i>sigmoid</i>	unnormalized sigmoid, no softmax	random	pretrained	Gu et al. [6]
<i>textinit</i>	softmax	pretrained SmolLM2-135M	pretrained	— (novel control)

The *glgate* and *sigmoid* levers have established sink effects in text-only models [20, 6]. *textinit* has no precedent there. It is an inheritance control for whatever sink structure text pretraining built into SmolLM2.

A scale confound in our gate variant. Unlike Qiu et al. [20], we initialize the G1 gate at exactly zero, so it opens at $\sigma(0) = 0.5$ and the arm begins as a half-scale output intervention as well as a gating one. We therefore write **Qiu-style G1 in our zero-initialized variant** throughout (§5).

Data and the two training regimes. The four-arm comparison trains on four curated subsets of `the_cauldron` [21], about 146K images, matched at about 100M tokens per arm. *textinit* stops at 60M, where its signatures have plateaued (§3.1). Reuse of that pool gives high visual-epoch counts, the objection the **RF** arm (random-fresh) answers (§3.2). RF re-trains the *baseline* recipe on a fresh FineVision stream [22] to 1B tokens, over about 4.6M natural images, at **2.39 effective visual epochs**. That is a **low-repetition** run rather than a repetition-free one, since examples repeat about 2.4 times on average. We estimate the overlap between the fresh pool and the repeated subsets at under 3%, from config-level composition rather than image-level deduplication. Swapping datasets trades the repetition confound for a domain-shift confound, which §5 takes up.

Three signatures, tracked separately. We log the three sink symptoms the text-LM literature reports together, each at its own granularity, following Gu et al. [6] at fixed sequence length. The decoder has $L = 30$ layers, $H = 9$ query heads and $G = 3$ KV heads per layer. We average every quantity below over valid query positions and over the fixed probe batch.

Concentration ($\text{Sink}_1^\varepsilon$) is the fraction of the $L \cdot H = 270$ (layer, query-head) pairs whose mean attention to position 0 exceeds ε . We use the $\varepsilon = 0.3$ default of [6] and check $\varepsilon \in \{0.2, 0.4\}$. Cross-arm tables report the stricter $\varepsilon = 0.2$, which makes an absence claim harder to pass.

Value-norm ratio (v-ratio) is the value norm at position 0 divided by the mean value norm over the other valid positions, per layer and averaged over the 30 layers. Below 1 is value-drain [4], above 1 amplification. Under grouped-query attention a value vector belongs to a KV group and repeats across 3 query heads, so the ratio rests on $L \cdot G = 90$ distinct value projections, not 270 (§3.3).

Residual-norm ratio (h-ratio) is the residual-stream norm at position 0 divided by the mean norm over the other positions, again per layer and averaged over the 30 layers. We call it a **massive-activation proxy**, because the literature defines massive activations by channel-level outliers [2, 5], which we never measured (§5).

All three metrics **anchor on position 0 by construction**, and we check that choice. At seed 0, per-position attention mass makes position 0 the maximum-mass token in every arm (Appendix C), and §5 handles the remaining seed-level caveat. The *sigmoid* arm reports the row-normalized attention view, which keeps concentration comparable across arms, and we also log the raw sigmoid mass (Appendix D).

Why a sink is expected at all. Softmax forces every attention row to sum to one, so a head with nothing worth retrieving must still put its mass somewhere [6]. Our *sigmoid* arm removes that constraint and tests it directly.

Probe. The function `probe_sinks()` re-walks the decoder from the live module weights in eager mode, fp32, no gradients, independent of the training path (fused SDPA kernels, autocast, `torch.compile`). Every call validates its hidden states against the real forward pass to a relative error below 10^{-2} , so the probe cannot drift from what the trained model computes. The probe batch is fixed across all runs and seeds, which keeps every number comparable, and probes fire every 100 optimizer steps, dense enough to timestamp each signature’s first threshold crossing. Appendix F gives the probe batch, token accounting and validation protocol.

Recipe. AdamW with weight decay 0.1, following [6], gradient clip 1.0, cosine schedule with 3% warmup. Arms in a comparison differ only in the lever under test. Two seeds for *baseline* and *sigmoid*, three for *g1gate* and *textinit*, one for *RF*. Appendix F gives the learning rates, precision and batch size.

Validation losses, and what they license. Training stays healthy in all reported runs. At the matched 100M-token checkpoint the held-out losses are 1.182 for *baseline*, 1.133 for *g1gate*, 1.206 for *sigmoid* and 0.877 for *textinit*, and RF reaches 0.638 at 1B tokens (Appendix F). Two cautions. *textinit* starts from a pretrained text decoder, so its lower loss reflects unequal compe-

tence, not a lever effect. Only *baseline*, *g1gate* and *sigmoid* are equal-token, equal-initialization comparisons. And the repeated-data arms show a large train–validation asymmetry (val_seen near 0.44 against val_unseen near 1.18), the overfitting signal that motivates RF. RF has no seen split. The weaker statement its data supports is a held-out loss that ends at 0.638 with a negative fitted slope over the second half, individual evaluations fluctuating (§5). **We did not run MMStar or any other downstream benchmark on any arm**, so this paper makes no capability claim (§5).

3. Results

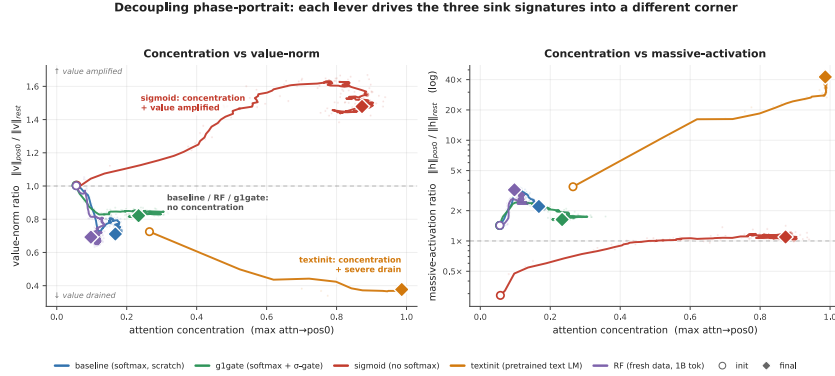


Figure 1: Decoupling phase-portrait. Every arm in concentration-against-value-norm space (left) and against the residual-norm ratio (right, log scale). Circles mark initialization, diamonds the final checkpoint (about 100M tokens, 60M for *textinit*, 1B for *RF*). Seed 0 unless labelled. **The horizontal axis is the maximum attention→pos0 over heads, not Table 1’s fraction above threshold**, so an arm can sit off zero here with $\text{Sink}^{\epsilon} = 0.000$. Paths smoothed, end markers raw (Appendix D). Were the three signatures one phenomenon, these trajectories would move along one direction.

3.1 Four training levers reach four different signature corners

Table 1 collapses the seeds into per-arm ranges, each arm at its final matched checkpoint: about 100M tokens, 60M for *textinit*, whose signatures have plateaued by then (per-seed table in Appendix B). **No two arms share a signature triple.**

Table 1: the four corners (n = 2–3 seeds per arm). Concentration is $\text{Sink}^{0.2}_1$, the fraction of the 270 (layer, query-head) pairs whose mean attention→pos0 exceeds 0.2. The v-ratio rests on 90 (layer, KV-head) value projections and the h-ratio on 30 layers (§2).

arm	concentration	value-norm	massive-activation proxy
baseline	absent (0.000)	mild drain (0.69–0.72)	moderate (1.7–2.2)
g1gate	near-absent (0.004–0.011)	milder drain (0.81–0.85)	moderate (1.7–2.2)
sigmoid	strong (0.76–0.83)	amplified (1.48–1.60)	no strong asymmetry (1.1–1.3)
textinit	strong (0.56–0.85)	strong drain (0.38–0.63)	extreme (5.5–42.5)

The value-norm axis alone takes three directions: drained hard, drained mildly, amplified. The corners separate pairwise on single axes. *g1gate* differs from *baseline* on the value-norm axis alone. Concentration is absent or near-absent and the residual-norm ratio moderate in both, while the gate makes the baseline’s mild drain milder still, a 15–19% drain rather than none. *sigmoid* and *textinit* both reach strong concentration and then part on the *direction* of the value-norm move and on the residual-norm ratio. The *sigmoid* corner is relative. Its concentration is row-normalized over a shrinking raw gate budget, with top-head raw pos0 mass 0.065 at the final checkpoint (Appendix D). The arms are not a factorial design, so these are intervention-associated profiles, not isolated causal effects of single levers. The trajectories in Figure 1 separate early and do not share one origin. *textinit* starts from a pretrained text decoder that already carries a subthreshold first-position bias, with $\text{Sink}^{0.3}_1$ still 0.000 at step 0.

Reproducibility differs by arm. *glgate* is tightest, $\text{Sink}^{0.2}_1$ of 0.004, 0.011 and 0.0037 across three seeds. The corner of *textinit* repeats across seeds and its magnitudes do not. Seed 0 sits far above the other two on every signature at once, h-ratio 42.5 against 5.5–12.2, partly because at seeds 1 and 2 its peak signatures move off position 0, where our metrics anchor (Appendix B, H). We therefore report *textinit*’s massive-activation proxy as a **range (5.5–42.5×) with a median near 12×** throughout, and treat the corner as the reproducible claim rather than any magnitude.

The corners also separate in time (Appendix H, Fig. A4, seed 0). The random-decoder softmax arms, *baseline*, *glgate* and *RF*, cross the norm thresholds without ever crossing concentration. *sigmoid* is the mirror image, crossing concentration without either norm threshold. *textinit* starts above both norm thresholds and crosses concentration later. Where a run crosses both kinds, the norm signatures come first, and under text initialization the three need not even settle on the same token. Appendix H gives the timings, the positional scan and the entropy-collapse correlate, and Appendix E sets out, as untested hypotheses, how each lever might move a different axis.

Figure 2 shows the separation inside a single head. Its rightmost panel carries the most weight. **The *textinit* stripe is already there at step 0**, imported with the text-LM weights before the model has seen one image.

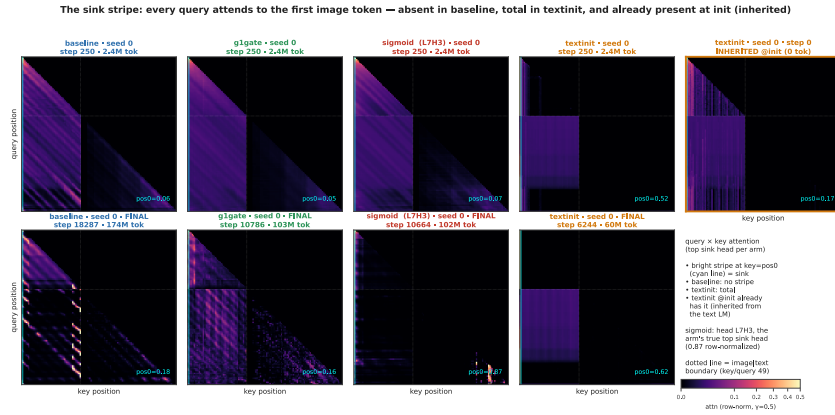


Figure 2: The sink stripe. Query $\times$ key attention of each arm’s top sink head, row-normalized, seed 0. Top row is early training (step 250, about 2.4M tokens), bottom row the final checkpoint. The cyan line marks key = pos0, the dotted line the image-to-text boundary. The stripe is *absent* in baseline throughout, strong in *textinit* (attn $\rightarrow$ pos0 = 0.62 at its final checkpoint), and *already present* at step 0 in *textinit* (rightmost panel), inherited from the text LM. The sigmoid column’s heat maps come from an auxiliary streaming batch, while every printed sigmoid number comes from the fixed probe batch (Appendix D).

3.2 The proxy grows across 1B low-repetition tokens with concentration at zero

The four-arm comparison reuses a 146K-image pool, so a reader could read its “concentration never emerges” result as an overfitting artifact. The RF arm answers that. It runs the identical *baseline* recipe on a fresh FineVision stream to one billion tokens at **2.39 effective visual epochs**, a **low-repetition** run rather than a repetition-free one, with a held-out loss that ends at 0.638 on a negative fitted slope over the second half (§2, §5).

Concentration never arrives. $\text{Sink}^{0.3}_1 = 0.000$ across the entire 0 $\rightarrow$ 1B run, and no head of the 270 crosses the threshold at any of about 700 probes.

Table 2: RF (fresh stream, n = 1 seed), init $\rightarrow$ 1B tokens.

signature	@ init	@ 1B	net
$\text{Sink}^{0.3}_1$ (concentration)	0.000	0.000	flat zero, entire run
max attn $\rightarrow$ pos0	0.056	0.098	$\approx$ flat, far below threshold
h-ratio (massive-activation proxy)	1.43	3.22	$\approx$ 2.3× , positive long-horizon trend
v-ratio (value-norm)	1.00	0.69	net drain, non-monotone

Warmup does not explain the rise. The h-ratio climbs from 2.40 at about 57M tokens to 3.22 at 1B, long after the 3% warmup window closes. The climb is not monotonic at probe resolution, so we call it a positive long-horizon trend. In Figure 1, right, the violet trajectory climbs and never moves rightward. The v-ratio ends below its initialization value, but about 75% of that drop happens in the first 57M tokens and part then recovers, so we report it as supporting context rather than ongoing emergence.

3.3 No consistent-sign head-level concentration–value-norm relationship

A tight coupling between concentration and value-drain should at minimum hold the sign of their per-head correlation constant across regimes. It does not. Table 3 gives the Pearson r between $\text{attn} \rightarrow \text{pos0}$ and value-norm ratio at each arm’s final checkpoint, seed 0, over the 90 (layer, KV-group) observations. We average a group’s query-head attention rather than triplicate its value observation (§2, scatter in Fig. A2).

Table 3: $r(\text{attn} \rightarrow \text{pos0}, \text{v-ratio})$, final checkpoint, seed 0.

baseline	g1gate	sigmoid	textinit	RF	pooled
+0.76	+0.57	−0.03	−0.79	+0.51	−0.20

These correlations are **descriptive, not inferential**. Heads within a layer are not independent, and each arm rests on a single seed here, so a p-value would mislead and we report none. Collapsing to the 90 KV groups removes the pseudoreplication a per-query-head reading would introduce, and the picture does not change (Appendix D).

The pattern is about **sign**, which pseudoreplication does not manufacture. Heads attending more to position 0 have *larger* value norms in the baseline regime and *smaller* ones under text initialization, and the pooled correlation is weak only because arms of opposite sign cancel. We claim that no consistent-sign relationship holds across arms, not that no coupling law exists. A fixed coupling would show one sign everywhere, and it does not.

4. Related work

Attention sinks and their companions in text language models. Gu et al. [6] give the canonical account: the sink token acts “more like key biases, storing extra attention scores,” carries small key and value norms as part of the same phenomenon, and connects to massive residual-stream activations [2, 5]. They also show unnormalized sigmoid attention prevents sink formation in text models up to 1B parameters, which our *sigmoid* arm builds on. Guo et al. [4] call the concentration and value-drain coupling “active-dormant” heads. Queipo-de-Llano, Arroyo et al. [7] make the strongest unity claim, and the only causal one. Ablating a model’s layer-0 massive activation removes both compression valleys and sink formation (§1). We reserve “causally unified” for that result alone. Peng et al. [24] trace a first-position sink circuit emerging early in from-scratch text pretraining, without separating the signatures.

Prior decoupling results: text-only, two axes. Two papers already separate pairs of these signatures, and we scope our claim around them. Sun, Canziani, LeCun and Zhu [9] show massive activations and attention sinks are dissociable architectural artifacts. A change of normalization scheme crushes the massive-activation spike while the sink ratio survives. Chen and Yao [10] decouple the same pair from the opposite direction and from scratch. A value-scale intervention in 0.1–0.3B text models keeps the sinks and suppresses massive activations. Neither treats value-norm drain as a third axis, and neither is multimodal. Fesser et al. [23] give the closest external support for tracking the value norm separately: one sink pattern can hide an “adaptive nop” (negligible value norms) or a “broadcast” that redistributes global information (low-rank outputs), though that work diagnoses trained vision transformers rather than tracking emergence. Qiu et al. [8] argue the opposite case, that outlier-driven rescaling by attention and residual sinks is essential to stable training. The field has settled neither question.

Multimodal sinks: mostly frozen backbones, inference time. Luo et al. [11] separate ViT-propagated from LLM-emerged sinks. Their Appendix A.4 tracks sink-dimension magnitudes across

alignment checkpoints, so a training-time view has precedent, though on a frozen vision transformer with a pretrained language model, and follows one magnitude rather than three signatures. Choi et al. [12] likewise distinguish vision-sinks from language-sinks in a frozen model and gate them by layer. Both establish distinct vision-side and language-side origins, which our *textinit* inheritance result fits. Neither gives dense, joint tracking of the three quantities in a decoder that starts from random weights. Vision transformers also grow high-norm “register” tokens of their own [25], hence the pretrained- encoder limitation in §5. A separate line ties these signatures to hallucination and grounding failure in deployed models and intervenes at inference time [13–16], the practical reason it matters which signature a mitigation moves.

The gating lever. Qiu et al. [20] introduce the head-specific elementwise sigmoid gate on attention output that our *glgate* arm adapts. In text models it “largely reduces the attention score allocated to the first token and decreases massive activations” while improving quality. Our zero-initialized variant confounds gating with initial output scaling (§2, §5). With that caveat, concentration is already absent in our baseline, so the gated arm differs on the value-norm axis. The drain becomes milder, 0.69–0.72 to 0.81–0.85, rather than disappearing, a difference invisible unless the three signatures are logged separately.

Positioning. The closest prior work separates at most two of the three axes, in text-only models, and the multimodal studies work mostly on frozen ones. Our claim is the conjunction. We track **concentration, value-norm drain and the residual-norm ratio jointly and separately in multimodal pretraining with a randomly initialized decoder**, which adds value-norm drain as a third axis beyond the text-only dissociations [9, 10]. Decoupling itself is not ours, and prior multimodal work could study emergence if it chose to.

5. Limitations

Metrics anchored on position 0. We measure all three signatures at the first image token, and check that anchor by per-position attention *mass* rather than *argmax* alone. Position 0 is the maximum-mass token in every arm at seed 0, and it stays so for *baseline*, *glgate* and *sigmoid* at every seed we scanned. It does **not** in *textinit*, where at seeds 1 and 2 the peaks move to other positions (per-position table in Appendix C). The pos0-anchored *textinit* magnitudes at those seeds therefore *understate* its peaks, one more reason to treat that arm’s corner rather than its magnitudes as the claim. In *RF* the attention *argmax* sits at pos1, but with mass 0.053 against 0.044 at pos0, a diffuse profile rather than a displaced sink, so the RF result does not depend on the anchor.

The gate arm carries a scale confound. We initialize the G1 gate at exactly zero, so it opens at $\sigma(0) = 0.5$ and halves attention output at step 0, where Qiu et al. [20] use ordinary initialization (§2). Gating and initial output scaling are therefore confounded in *glgate*. We cannot attribute its differences from baseline to gating alone, and a scale-matched control is future work.

Pretrained vision encoder: no arm is fully from scratch. The SigLIP encoder is pretrained and trainable in every arm, and *textinit* also uses a pretrained decoder. We study vision–language pretraining with randomly initialized decoders, not from-scratch training of the whole model. Vision transformers grow high-norm register tokens of their own [25], and sinks can propagate from a vision transformer into a vision–language model [11], so part of our residual-norm signal could be inherited rather than decoder-formed. Our defense is the trajectory. The h-ratio starts at 1.0–1.4 and *rises* to 3.22 across 1B tokens in RF, where inheritance from a static encoder predicts a high, flat h-ratio from step 0. A randomly initialized vision transformer, which would isolate the decoder, is future work.

The h-ratio is a proxy, not a measurement of massive activations. The literature defines massive activations by channel-level outliers [2, 5]. We measured a position-specific residual-norm ratio and never computed channel-level statistics (§2). A large h-ratio is consistent with massive activations without establishing them.

Token scale. Our runs reach at most 1B tokens per arm, against the roughly 5B canonical in the text-LM sink literature [6]. Text-LM sinks form near step 1000 [7], far inside our range, but a sig-

nature absent at 1B could still emerge later.

Reproducibility of textinit magnitudes. The massive-activation proxy of *textinit* is seed-sensitive, an h-ratio of 5.5–42.5 across three seeds, with seed 0 the outlier on every signature. The reproducible claim is the corner: strong concentration, strong drain, large residual-norm ratio. No specific magnitude is.

Provenance and seed count. The seed-0 raw probes for the four-arm comparison come from a checksummed archive summary rather than first-hand re-derivation. Seeds 1 and 2 we *did* re-derive independently, an audit that caught a metric-labeling error in an earlier internal consolidation (Appendix G). *baseline* and *sigmoid* have two seeds, *lgate* and *textinit* three, RF a **single seed**. RF also contains one **weights-only optimizer restart at about 57M tokens**, forced by an out-of-memory error. We reloaded the weights and discarded the AdamW moment estimates, so RF is not one uninterrupted optimizer trajectory. The audit verified continuity across that seam: v-ratio and h-ratio identical at the shared checkpoint, a double-covered 600-step overlap diverging only within probe noise, concentration 0.000 on both sides. Concentration was reproducibly zero across both repeated-data baseline seeds, adequate support for the negative claim, and a second fresh-data seed would strengthen it.

What the RF control does and does not establish. RF has no distinct seen split, so we cannot run for it the `val_seen / val_unseen` comparison that exposes memorization in the repeated arms. The weaker statement its data supports is the falling-slope statement of §2. Its streaming shuffle buffer dropped from 1500 to 500 examples partway through, again for memory, so stream ordering is not homogeneous, though the $\text{Sink}^{0.3}_1 = 0.000$ result and the h-ratio rise hold within each regime separately. Its probe batch is still the **fixed repeated-the_cauldron tail** used by the other arms. That keeps RF’s signatures comparable to theirs, which is what the negative result needs, at the cost of measuring them on data from the other distribution. And RF buys lower repetition by changing dataset, so it trades the repetition confound for a domain-shift one, and we chose the fresh pool to minimize shift. The under-3% overlap figure is config-level, not image-level (§2), and the third, domain-matched control we did not run is the known follow-up.

We report no benchmark accuracy. We measure sink signatures only (§2): no downstream benchmark evaluation, such as MMStar, on any arm, and no claim about how signature dissociation relates to downstream capability.

6. Conclusion

The coupling is lever-dependent. Across four training levers, the three signatures land in four distinct corners: value norms are strongly drained, mildly drained, or amplified, and on a billion fresh tokens the massive-activation proxy more than doubles while concentration stays at zero. This extends two-way dissociations in text-only models [9, 10] to a third, separately measured axis in multimodal pretraining. In text LMs, massive activations can causally produce sinks [7]; our results show the coupling can also fail to form.

The practical implication is simple: across our arms, no one signature was a proxy for the others. A model without a concentration sink can still develop a growing residual-norm asymmetry, and an intervention judged on one signature may leave the others unchanged.

Next steps. A randomly initialized vision encoder, to isolate what the decoder contributes to the residual-norm signal. A fresh-data run past 1B tokens. A scale-matched gate control, to separate gating from the half-scale confound of §2.

Reproducibility. A self-validating probe (§2) computes all signatures on a fixed probe batch, from logs taken every 100 steps, and the Appendix holds the per-seed tables. We will release the code, the probe, the run configurations, the per-run logs, and the training checkpoints on acceptance. We withhold the links for double-blind review.